SIT723 Research Techniques & Applications

Task 2.1p Summarise Literature

This document summarises 40 research papers related to my project on evaluating the stability of large language models using semantic entropy. The selected papers cover topics such as prompt sensitivity, semantic consistency, adversarial robustness, uncertainty estimation, and biomedical applications. From reviewing these papers, it becomes clear that even though LLMs often achieve high accuracy, they can still produce inconsistent outputs when the input is slightly changed. Also, highlights limitations in traditional evaluation methods and shows the need for better approaches that focus on meaning rather than just correctness. The papers are given bellow:

Paper Details	Summary	Key Points
L. Kuhn, Y. Gal, and S. Farquhar, 2023, "Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation," 11th International Conference on Learning Representations (ICLR), DOI / Identifier: openreview.net/forum?id=VD-AYtP0dve	This paper introduces semantic entropy as an uncertainty measure for natural language generation. Instead of treating different wordings as completely different outputs, it groups responses by shared meaning and then computes entropy in meaning-space. The paper shows that this approach is better than token-level uncertainty measures like predictive entropy and p(True), especially on question answering tasks such as TriviaQA and CoQA. For this project, it is the most important paper because it provides the core method for evaluating semantic stability under meaning-preserving input changes.	First formal definition of semantic entropy for NLG uncertainty. Outperforms token-level uncertainty baselines. Model-agnostic and does not require changing the model. Directly supports semantic stability evaluation.
S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, 2024, "Detecting hallucinations in large language models using semantic entropy," Nature, DOI: 10.1038/s41586-024-07421-0	This paper extends the semantic entropy idea into hallucination detection. The authors show that when a model gives semantically diverse answers to the same question, the response is more likely to be wrong or fabricated. A major strength of the paper is that it does not rely on labelled hallucination data, which makes it more practical across tasks. This paper is very	Shows semantic entropy can detect factual hallucinations. Validates the method in a top-tier journal. Uses bidirectional entailment clustering for semantic equivalence. Strong evidence that

	relevant because it validates semantic entropy not only as a theoretical idea but also as a practical reliability signal for LLMs.	instability and unreliability are closely linked.
Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, D. Chen, W. Dai, A. Madotto, and P. Fung, 2023, "Survey of hallucination in natural language generation," ACM Computing Surveys, DOI: 10.1145/3571730.	This survey gives a broad overview of hallucination in natural language generation and explains that hallucinations are not all the same. It distinguishes between intrinsic hallucinations, where the model contradicts available information, and extrinsic hallucinations, where it generates unsupported content. The paper also reviews causes connected to data, model design, and inference strategies. For this study, it helps explain why semantic instability matters, because unstable outputs are often connected to hallucinated or unreliable responses.	Foundational hallucination taxonomy. Distinguishes intrinsic and extrinsic hallucinations. Reviews root causes across data, model, and inference. Useful background for reliability-focused evaluation.
L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, 2025, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," ACM Transactions on Information Systems, DOI: 10.1145/3703155.	This paper updates hallucination research specifically for the LLM era. It separates factuality hallucinations from faithfulness hallucinations and explains why instruction-following models need a more detailed taxonomy than earlier NLG systems. It also reviews mitigation strategies such as retrieval, factchecking, and reinforcement methods. For this project, the most useful point is that reliable behavior should include consistency across semantically equivalent inputs, not just high one-shot accuracy.	Extends hallucination research for modern LLMs. Separates factuality and faithfulness errors. Reviews detection and mitigation methods. Reinforces the importance of semantic consistency in evaluation.
M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr, 2024, "Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt	This paper demonstrates that language models can show very large performance changes even when prompts are only changed in superficial, meaning-preserving ways. The	Shows major performance swings from prompt formatting alone. Demonstrates that scale and instruction

formatting," 12th International Conference on Learning Representations (ICLR), DOI / Identifier: arXiv:2310.11324	authors report variations of up to 76 percentage points in some few-shot settings, which is a very strong result. This means benchmark accuracy can be misleading if the prompt format changes slightly. In relation to this project, the paper strongly motivates why semantic stability should be measured directly rather than assumed from accuracy scores.	tuning do not fully solve prompt sensitivity. Supports the argument that accuracy can hide instability. Important motivation for semantic entropy.
J. Zhuo, S. Zhang, X. Fang, H. Duan, D. Lin, and K. Chen, 2024, "ProSA: Assessing and understanding the prompt sensitivity of LLMs," Findings of the Association for Computational Linguistics: EMNLP 2024, DOI / Identifier: ACL Anthology: 2024.findings-emnlp.108	This paper proposes the ProSA framework and the PromptSensiScore metric for measuring prompt sensitivity in a more systematic way. It shows that prompt sensitivity is not uniform across tasks and that few-shot examples can partially reduce the problem. The paper also links prompt robustness to decoding confidence, which is interesting because it suggests that confidence and stability are related but not identical. This is useful for the current study because it helps explain why some prompts trigger more stable behaviour than others.	Introduces PromptSensiScore for prompt sensitivity. Shows sensitivity differs across datasets. Finding few-shot examples can reduce sensitivity. Suggests confidence is related to, but not the same as, robustness.
X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, 2023, "Self-consistency improves chain of thought reasoning in language models," 11th International Conference on Learning Representations (ICLR), DOI / Identifier: arXiv:2203.11171	This paper introduces self-consistency decoding, where the model generates multiple reasoning paths and the final answer is selected by majority agreement rather than a single greedy output. The results show clear improvements across arithmetic and commonsense reasoning benchmarks. Although this is mainly presented as a decoding strategy, it is highly relevant to semantic entropy because both approaches rely on agreement or stability across multiple outputs. The paper therefore supports the idea that stable	Uses agreement across multiple outputs to improve reasoning. Show performance gains on several benchmarks. Connects output stability with correctness. Conceptually linked to semantic entropy.

	outputs are often associated with more reliable answers.	
H. Raj, V. Gupta, D. Rosati, and S. Majumdar, 2023, “Semantic consistency for assuring reliability of large language models,” arXiv preprint, DOI / Identifier: arXiv:2308.09138.	This paper focuses directly on semantic consistency and shows that large language models often produce different answers to semantically equivalent questions. The authors also propose the Ask-to-Choose prompting method to improve consistency without supervised retraining. What makes this paper important is that it clearly shows a gap between being correct once and being consistently correct across paraphrases. That gap is exactly the problem that the current semantic entropy project is trying to measure more formally.	Direct evidence that accuracy and consistency are different. Introduces Ask-to-Choose as a consistency improvement method. Supports the need for stability-based evaluation. Very closely aligned with this project’s core idea.
S. Goyal, S. Doddapaneni, M. M. Khapra, and B. Ravindran, 2023, “A survey of adversarial defenses and robustness in NLP,” ACM Computing Surveys, DOI: 10.1145/3593042	This survey covers adversarial attacks and defense strategies in NLP, including character-level, word-level, and sentence-level perturbations. One important contribution is its distinction between semantic quality and attack success rate, because an attack can be effective while still changing the original meaning. For this project, that distinction matters a lot because the goal is to test meaning-preserving perturbations rather than arbitrary adversarial noise. The survey therefore helps define a more careful perturbation methodology.	Comprehensive overview of NLP adversarial robustness. Separates semantic preservation from attack success. Helps justify using meaning-preserving perturbations. Useful for methodology design.
S. S. Alahmari, L. Hall, P. R. Mouton, and D. Goldgof, 2025, “Large language models robustness against perturbation,” Scientific Reports, DOI: 10.1038/s41598-025-29770-0	This paper evaluates a group of contemporary LLMs under perturbations such as keyboard noise and word replacement. The results show that larger models usually preserve meaning better than smaller ones, but instability remains even in stronger models. This finding is useful because it	Compares multiple LLMs under perturbation. Finding larger models is more robust but still imperfect. Shows semantic instability persists in advanced

	suggests that model scale alone does not solve robustness problems. In the context of this project, it supports the idea that semantic stability needs to be measured explicitly rather than assumed to improve automatically with model size.	systems. Relevant to model comparison.
X. Xu, K. Kong, N. Liu, L. Cui, D. Wang, J. Zhang, and M. Kankanhalli, 2024, "An LLM can fool itself: A prompt-based adversarial attack," 12th International Conference on Learning Representations (ICLR), DOI / Identifier: OpenReview: VVgGbB9TNV.	This paper presents PromptAttack, where an adversarial LLM is used to generate perturbations that trick another LLM. A striking point is that the target model can effectively be manipulated through prompt design alone, without changing its parameters. The paper shows that attack success rates can increase significantly on benchmark tasks, which suggests that prompt sensitivity is not just a surface issue but reflects deeper representational instability. For this project, that makes it a strong supporting paper for testing robustness under controlled, meaning-preserving changes.	Shows LLMs can generate adversarial perturbations through prompting. Suggests instability are partly intrinsic to model representation. Connects prompt sensitivity with robustness problems. Supports perturbation-based evaluation.
Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, 2021, "Domain-specific language model pretraining for biomedical natural language processing," ACM Transactions on Computing for Healthcare, DOI: 10.1145/3458754	This paper introduces PubMedBERT and argues that pretraining from scratch on purely biomedical text leads to better performance than continuing from a general-domain model. The authors show improvements across biomedical benchmarks and suggest that in-domain vocabulary and data distribution matter a lot. For this project, it is an important reference because it justifies comparing domain-specific biomedical models against more general LLMs when testing semantic stability in medical text.	PubMedBERT. Shows domain-specific pretraining improves biomedical NLP. Supports domain-specific model comparison in this study.
R. Tinn, H. Cheng, Y. Gu, N. Usuyama, X. Liu, T. Naumann, J.	This paper studies fine-tuning stability in biomedical NLP	Shows PubMedBERT has

Gao, and H. Poon, 2023, "Fine-tuning large neural language models for biomedical natural language processing," Patterns (Cell Press), DOI: 10.1016/j.patter.2023.100729.	and shows that PubMedBERT is not only strong in performance but also more stable than several other BERT-style models. It further discusses strategies such as layer freezing and optimisation choices that can improve low-resource performance. This matters for the current project because stability is not only about outputs under paraphrase, but also about how robustly a model adapts to biomedical tasks.	strong fine-tuning stability. Links domain vocabulary with improved robustness. Useful for understanding biomedical model behaviour beyond raw accuracy.
J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, 2020, "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, DOI: 10.1093/bioinformatics/btz682	This is one of the foundational papers in biomedical NLP. BioBERT is introduced as a model pretrained on PubMed and PMC data, and it improves several biomedical tasks such as named entity recognition, relation extraction, and question answering. For this project, BioBERT serves as a key baseline because it represents the earlier generation of biomedical language models that can be compared with newer approaches like PubMedBERT and BioMistral.	Foundational biomedical transformer model. Improves multiple biomedical NLP tasks. Strong comparison baseline for domain-specific evaluation.
Y. Labrak, A. Bazoge, E. Morin, P.-A. Gourraud, M. Rouvier, and R. Dufour, 2024, "BioMistral: A collection of open-source pretrained large language models for medical domains," Findings of the Association for Computational Linguistics: ACL 2024, DOI / Identifier: ACL Anthology: 2024.findings-acl.348.	This paper introduces BioMistral, a family of medical-domain LLMs built from Mistral-7B and adapted on biomedical corpora. It shows that open-source models can perform strongly on medical QA tasks, sometimes even outperforming larger proprietary systems like GPT-3.5. This is especially useful for the current project because it provides a realistic biomedical LLM comparison model for semantic entropy experiments.	Open-source medical LLM family. Strong medical QA performance at moderate scale. Important candidate model for biomedical stability comparison.
K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung,	This paper presents Med-PaLM and shows that LLMs can	Demonstrates strong clinical knowledge

N. Scales et al., 2023, “Large language models encode clinical knowledge,” Nature, DOI: 10.1038/s41586-023-06291-2	perform very strongly on medical questions, answering benchmarks, approaching expert-level results in some settings. At the same time, it also highlights ongoing issues with uncertainty communication, harm, and bias. This makes the paper very important for the current study because it shows that high medical accuracy does not automatically mean safe or stable understanding.	in LLMs. Highlights remaining reliability and uncertainty issues. Supports the argument that benchmark performance alone is not enough.
P. Hager, F. Jungmann, R. Holland, K. Bhatt, E. Butcher, P. Bhatt, and D. Rückert, 2024, “Evaluation and mitigation of the limitations of large language models in clinical decision-making,” Nature Medicine, DOI: 10.1038/s41591-024-03097-1	This paper tests LLMs on realistic clinical tasks rather than standard benchmark questions and finds that they perform much worse than benchmark scores might suggest. The models struggle with guideline adherence, laboratory interpretation, and clinical reasoning. This is highly relevant to the project because it clearly demonstrates the “evaluation illusion,” where good benchmark accuracy gives a misleading view of real-world reliability.	Strong evidence for the evaluation illusion in medicine. Shows failure in realistic clinical decision-making. Reinforces the need for better reliability metrics.
M. Afshar, Y. Gao, D. Gupta, E. Croxford, and D. Demner-Fushman, 2024, “On the role of the UMLS in supporting diagnosis generation proposed by large language models,” Journal of Biomedical Informatics, DOI: 10.1016/j.jbi.2024.104707.	This paper investigates whether UMLS grounding can improve LLM diagnosis generation and evaluation. It finds that UMLS-based knowledge paths improve results and that UMLS-grounded metrics align better with expert judgement than common overlap-based metrics like ROUGE. For this study, the paper is especially important because it supports using UMLS concept identifiers as a formal basis for semantic equivalence clustering.	Shows UMLS improves both diagnosis quality and evaluation alignment. Supports CUI-based semantic clustering. Directly relevant to concept-level semantic entropy.
N. J. Dobbins, 2025, “Generalizable and scalable multistage biomedical concept normalization leveraging large	This paper studies LLMs in biomedical concept normalization and shows that LLMs by themselves perform	Shows direct LLM concept normalization is weak. Supports

language models," Research Synthesis Methods, DOI / Identifier: 10.1017/rsm.2025.9 / PMC12527512.	poorly when mapping phrases directly to UMLS concepts. However, performance improves when they are placed inside a structured multi-stage pipeline. This is highly relevant because it suggests that variability in concept assignment is a real and measurable problem, which is exactly what concept-level semantic entropy is designed to capture.	structured, concept-level evaluation. Strong motivation for measuring instability in CUI assignments.
T. Savage, J. Wang, R. Gallo, A. Boukil, V. Patel, S. A. A. Safavi-Naini, A. Soroush, and J. H. Chen, 2024, "Large language model uncertainty proxies: Discrimination and calibration for medical diagnosis and treatment," Journal of the American Medical Informatics Association, DOI: 10.1093/jamia/ocae254.	This paper evaluates uncertainty proxies for medical LLM tasks using output similarity and embedding-based consistency measures. It finds that consistency-style signals can help distinguish correct from incorrect answers, although calibration is still not perfect. This matters for the current project because it shows that consistency-based uncertainty is useful in medical settings and can complement semantic entropy.	Apply consistency-based uncertainty measures in medicine. It shows some discrimination between correct and incorrect outputs. Useful comparison points for entropy-based uncertainty.
M. Agrawal, I. Y. Chen, F. Gulamali, and S. Joshi, 2025, "The evaluation illusion of large language models in medicine," npj Digital Medicine, DOI: 10.1038/s41746-025-01963-x.	This paper argues that the apparent success of medical LLMs is often exaggerated by benchmark design, task selection, and evaluation metrics. The authors show that common automated scores do not align well with expert judgement, especially in clinically meaningful settings. This paper is important because it directly supports the need for stronger, more principled evaluation methods such as semantic entropy.	Strong critique of current medical LLM evaluation. Shows mismatch between benchmarks and real clinical usefulness. Supports better reliability-focused metrics.
S. Zhou, Y. Cheng, W. Zhang, X. Liang, C. Li, Y. Li, Y. Hu, and J. Liu, 2025, "Uncertainty-aware large language models for explainable disease diagnosis," npj Digital Medicine, DOI: 10.1038/s41746-025-02071-6.	This paper introduces ConfiDx, an uncertainty-aware diagnosis framework that includes explicit uncertainty labels during fine-tuning. The study shows that modelling diagnostic uncertainty can	Demonstrates practical value of uncertainty modelling. Reduces misdiagnosis in ambiguous cases. Complements

	reduce misdiagnosis when clinical evidence is incomplete. For this project, it is useful because it shows uncertainty is not just a theoretical issue but something with clear clinical consequences.	unsupervised semantic entropy with a supervised uncertainty approach.
S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, Z. Hatfield-Dodds, N. DasSarma, E. Tran-Johnson et al., 2022, “Language models (mostly) know what they know,” arXiv, DOI / Identifier: arXiv:2207.05221	This paper asks whether language models can estimate their own correctness using p(True). The authors find that while the models may look reasonably calibrated on average, they are often unreliable at the individual instance level. This is very relevant to the current project because it highlights that self-reported confidence is not enough, which strengthens the case for external metrics such as semantic entropy.	Shows internal confidence is imperfect. Distinguishes aggregate calibration from instance-level reliability. Supports the need for external uncertainty metrics.
J. Novikova, C. Anderson, B. Blili-Hamelin, D. Rosati, and S. Majumdar, 2025, “Consistency in language models: Current landscape, challenges, and future directions,” arXiv preprint, DOI / Identifier: arXiv:2505.00268	This survey reviews multiple forms of consistency in language models, including hypothetical, compositional, factual, and moral consistency. It shows that models perform especially poorly on hypothetical and compositional consistency, and it also points out that most consistency research is concentrated on English text-to-text models. This is useful because it shows that consistency is a broad research area and that semantic stability under paraphrase is one important part of it.	Presents consistency as a multi-dimensional concept. Shows major gaps in hypothetical and compositional consistency. Help position this study within the wider consistency of literature.
D. Jin, Z. Jin, J. T. Zhou, and P. Szolovits, 2020, “Is BERT really robust? A strong baseline for natural language attack on text classification and entailment,” Proceedings of the AAAI Conference on Artificial Intelligence, DOI: 10.1609/aaai.v34i05.6311	This paper introduces TextFooler, a widely cited adversarial attack that uses synonym substitution to flip model predictions while preserving human-judged meaning. The results show that even strong NLP models can be highly fragile under small wording changes. For this project, the paper is important	Introducing TextFooler. Shows strong vulnerability to synonym substitution. Important precedent for perturbation-based stability testing.

	because it demonstrates why meaning-preserving perturbations are an effective way to probe model understanding.	
J. Ebrahimi, A. Rao, D. Lowd, and D. Dou, 2018, "HotFlip: White-box adversarial examples for text classification," Proceedings of the 56th Annual Meeting of the ACL (Short Papers), DOI: 10.18653/v1/P18-2006	This paper introduces HotFlip, a gradient-based adversarial method that changes characters or words to reduce model accuracy. Even very small modifications can cause large behavioural shifts. Although some attacks may not preserve meaning as cleanly as synonym substitution, the paper is still useful because it shows how fragile text models can be under minimal changes.	Early influential adversarial NLP paper. Shows large effects from small edits. Useful background for robust literature.
P. Rajpurkar, R. Jia, and P. Liang, 2018, "Know what you don't know: Unanswerable questions for SQuAD," Proceedings of the 56th Annual Meeting of the ACL, DOI: 10.18653/v1/P18-2124	This paper introduces SQuAD 2.0, which combines answerable and unanswerable questions to test both extraction ability and abstention. It is a stronger benchmark than earlier extractive QA datasets because it requires models to recognise when evidence is insufficient. In this project, SQuAD 2.0 is relevant as a cross-domain validation dataset for studying semantic uncertainty outside the biomedical setting.	Introduces SQuAD 2.0 with unanswerable questions. Useful for testing uncertainty and abstention. Serves as cross-domain validation in this project.
S. Mohan and D. Li, 2019, "MedMentions: A large biomedical corpus annotated with UMLS concepts," Proceedings of the 2019 Conference on Automated Knowledge Base Construction (AKBC), DOI / Identifier: arXiv:1902.09476	This paper introduces MedMentions, one of the largest biomedical entity-linking datasets, with thousands of PubMed abstracts and more than 350,000 UMLS-linked mentions. The dataset is especially important because it provides a structured way to evaluate semantic equivalence at the concept level through CUIs. For this project, MedMentions is the main dataset that makes concept-level semantic entropy practical.	Large UMLS-linked biomedical dataset. Provides concept-level labels through CUIs. Core dataset for semantic entropy evaluation in the medical domain.
M. Neumann, D. King, I. Beltagy, and W. Ammar, 2019, "ScispaCy:	This paper presents scispaCy, a biomedical NLP toolkit that	Biomedical NLP toolkit with UMLS

Fast and robust models for biomedical natural language processing," Proceedings of the 18th BioNLP Workshop and Shared Task, ACL, DOI / Identifier: ACL Anthology: W19-5034	includes entity recognition and UMLS linking functionality. It is widely used because it is practical, efficient, and tailored for scientific and clinical text. In this project, scispaCy is an implementation-level reference because it provides the tools needed to map generated outputs to UMLS concepts.	linking. Practical tool for implementation. It is important for turning model output into CUI-based clusters.
C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, 2017, "On calibration of modern neural networks," Proceedings of the 34th International Conference on Machine Learning (ICML), DOI / Identifier: PMLR: proceedings.mlr.press/v70/guo17a	This paper is a foundational calibration study showing that modern neural networks are often overconfident even when they are wrong. It also proposes temperature scaling as a simple post-hoc calibration technique. For this project, the paper is important because it helps separate the idea of confidence calibration from semantic consistency; a model can be calibrated in a probabilistic sense but still unstable across semantically equivalent inputs.	Foundational calibration paper. Show confidence and correctness are not the same. Useful for distinguishing calibration from semantic stability.
Y. Geifman and R. El-Yaniv, 2017, "Selective Classification for Deep Neural Networks," Advances in Neural Information Processing Systems (NeurIPS 2017), DOI / Identifier: NeurIPS 2017 proceedings.	This paper introduces selective classification, where a model can abstain from making a prediction when confidence is low. The main idea is to improve reliability by allowing the system to reject uncertain cases instead of forcing a possibly wrong answer. For this project, the paper is not directly about semantic entropy, but it is relevant as a broader reliability framework because both approaches aim to make model output safer and more trustworthy under uncertainty.	Introduces abstention-based reliability. Shows uncertainty can be operationalised in decision-making. Useful as a broader uncertainty baseline.
G. G. Şahin, 2022, "To augment or not to augment? A comparative study on text augmentation techniques for low-resource NLP," Computational Linguistics, DOI: 10.1162/coli_a_00425.	This paper compares augmentation methods such as back-translation, synonym replacement, syntactic reordering, and noise injection across multiple languages and tasks. The authors conclude	Compares multiple text augmentation methods. Shows augmentation benefits are task- and language-dependent. Useful

	that there is no single universally best augmentation strategy. This is directly useful for the current study because perturbation generation needs to preserve meaning as much as possible, and this paper shows that some methods are safer than others depending on the task.	for choosing perturbation strategies carefully.
A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, 2023, “Large language models in medicine,” Nature Medicine, 29(8), pp. 1930–1940, DOI: 10.1038/s41591-023-02448-8.	This review provides a broad overview of LLM applications in medicine, including diagnosis support, clinical text generation, patient communication, and drug discovery. At the same time, it clearly highlights key risks such as hallucination, bias, and the mismatch between benchmark success and real-world trustworthiness. This makes it an important framing paper for the medical relevance of semantic stability evaluation.	Broad review of medical LLM applications and risks. Highlights hallucination as a major deployment concern. Supports the need for domain-specific reliability evaluation.
W. Boag, D. Doss, T. Naumann, and P. Szolovits, 2018, “What’s in a note? Unpacking predictive value in clinical note representations,” AMIA Joint Summits on Translational Science Proceedings, DOI / Identifier: PMC5961801.	This paper studies how different representations of clinical notes affect predictive performance. It shows that unstructured text contains useful clinical information that cannot be fully replaced by structured codes alone. For this project, the paper is a useful background reference because it reinforces the importance of meaning-rich clinical text and the need to evaluate how models handle it consistently.	Shows clinical text carries important predictive information. Highlights the value of unstructured notes. Useful context for medical text evaluation.
J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, 2022, “Chain-of-thought prompting elicits reasoning in large language models,” Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS), DOI / Identifier: ACM DL: 10.5555/3600270.3602070	This paper shows that chain-of-thought prompting can significantly improve performance on reasoning tasks by encouraging the model to generate intermediate steps. It is a landmark paper because it changed how reasoning in LLMs is studied. For this project, it matters because	Landmark paper in reasoning prompting. Shows step-by-step prompts improve performance. Provides context for self-consistency and stability research.

	chain-of-thought and self-consistency connect naturally to the idea that stable reasoning traces can support more reliable outputs.	
R. Moradi and N. Samwald, 2022, “Improving the robustness and accuracy of biomedical language models through adversarial training,” Journal of Biomedical Informatics, DOI: 10.1016/j.jbi.2022.104114	This paper applies adversarial training in biomedical NLP and finds that robustness and task accuracy can improve together. Studying is important because it shows that robustness is not only something to measure after the fact; it can also be improved during training. For this project, it provides a relevant biomedical baseline for comparing whether stable models can be developed, not just evaluated.	Shows adversarial training can improve both robustness and accuracy. Biomedical-specific robustness reference. Useful baseline for future robustness comparisons.
A. Névéol, H. Dalianis, S. Velupillai, G. Savova, and P. Zweigenbaum, 2018, “Clinical natural language processing in languages other than English: Opportunities and challenges,” Journal of Biomedical Informatics, DOI: 10.1016/j.jbi.2018.01.001	This review discusses the state of clinical NLP beyond English and shows that there are major gaps in corpora, annotation standards, and model availability in many languages. It is relevant to this project because it highlights an important scope limitation: most current semantic consistency and biomedical LLM research is still English-centred.	Highlights multilingual gaps in clinical NLP. Shows the field is heavily English-focused. Useful for acknowledging research scope limitations.
O. Bodenreider, 2004, “The Unified Medical Language System (UMLS): Integrating biomedical terminology,” Nucleic Acids Research, DOI: 10.1093/nar/gkh061	This foundational paper explains how the UMLS Metathesaurus integrates many biomedical vocabularies into a unified concept system using Concept Unique Identifiers. It also describes semantic types and supporting tools. For this project, this is one of the most important background references because UMLS CUIs are the basis for defining semantic equivalence when computing concept-level semantic entropy.	Foundational UMLS reference. Explains CUIs and semantic types. Provides the formal basis for concept-level equivalence in this study.
F. Zhu, W. Lei, C. Wang, J. Zheng, S. Poria, and T.-S. Chua, 2021, “Retrieving and reading: A	This survey reviews open-domain question answering systems, including retrieval,	Reviews the architecture of open-domain QA

comprehensive survey on open-domain question answering," arXiv preprint, DOI / Identifier: arXiv:2101.00774	reading comprehension, and generation approaches. It also covers major QA benchmarks such as SQuAD, TriviaQA, and Natural Questions. In the current project, this paper is useful because it gives broader context for why QA tasks are commonly used to study uncertainty and model understanding.	systems. Covers key QA benchmarks. Provides background for using SQuAD 2.0 as cross-domain validation.
P. Manakul, A. L. Liusie, and M. J. F. Gales, 2023, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), DOI / Identifier: ACL Anthology: 2023.emnlp-main.557.	This paper introduces SelfCheckGPT, a zero-resource method for detecting hallucinations in large language models by sampling multiple responses to the same prompt and checking their consistency. The key insight is that if a model is confident about a fact, repeated samples will tend to agree; if it is uncertain or hallucinating, the samples will diverge. The method supports multiple consistency signals including NLI-based entailment, n-gram overlap, and prompting-based checking, without requiring any external knowledge base or labelled data. For this project, SelfCheckGPT is useful as a direct comparison baseline because it represents an alternative approach to measuring output reliability through consistency, but without using a formal semantic equivalence criterion such as UMLS CUI clustering.	Zero-resource hallucination detection using output consistency. Does not require clustering or labelled data. Multiple consistency signals supported. Strong methodological comparison points for semantic entropy.